Sample Paper 19

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

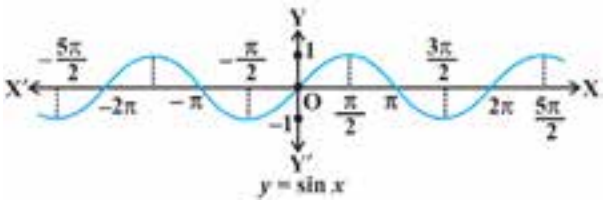
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \sin x$ intersect the origin in the given image?



(a) $x = -\pi$ (b) $x = 0$

(c) $x = \pi$ (d) $x = 2\pi$

2. Order of the differential equation of the family of all concentric circles centred at (h, k), is

(a) 2 (b) 3

(c) 1 (d) 4

3. If m and n are the order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^5 + 4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\frac{d^3y}{dx^3}} + \frac{d^3y}{dx^3} = x^2 - 1, \text{ then}$$

(a) $m = 3, n = 2$ (b) $m = 3, n = 3$

(c) $m = 3, n = 5$ (d) $m = 3, n = 1$

4. The vectors $\vec{a} = 2\hat{i} - 3\hat{j}$ and $\vec{b} = -4\hat{i} + 6\hat{j}$ are

(a) Coincident

(b) Parallel

(c) Perpendicular

(d) Neither parallel nor perpendicular

5. The area bounded by $y = \log x$, X-axis and ordinates $x = 1$, $x = 2$ is

(a) $\frac{1}{2}(\log 2)^2$ (b) $\log(2/e)$

(c) $\log(4/e)$ (d) $\log 4$

6. A straight line which makes an angle of 60° with each of y and z axes, inclined with x-axis at an angle of

(a) 30° (b) 45°

(c) 75° (d) 60°

7. It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P(\frac{A}{B}) = \frac{1}{2}$ and $P(\frac{B}{A}) = \frac{2}{3}$. Then, $P(B)$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

8. If $\sin^{-1}x = \theta + \beta$ and $\sin^{-1}y = \theta - \beta$, then $1 + xy$ is equal to

(a) $\sin^2\theta + \sin^2\beta$ (b) $\sin^2\theta + \cos^2\beta$

(c) $\cos^2\theta + \cos^2\beta$ (d) none of these

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9. Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}, \vec{b} = \hat{j}$. The value of $\vec{c}$ for which $\vec{a}, \vec{b}, \vec{c}$ form a right handed system is

(a) $\vec{0}$ (b) $z\hat{i} - x\vec{k}$

(c) $-z\hat{i} + x\hat{k}$ (d) $y\hat{j}$

10. $\int \frac{\sec^2(\sin^{-1}x)}{\sqrt{1-x^2}}dx$ is equal to

(a) $\sin(\tan^{-1}x) + C$ (b) $\tan(\sec^{-1}x) + C$

(c) $\tan(\sin^{-1}x) + C$ (d) $-\tan(\cos^{-1}x) + C$

11. If R is a relation on the set N, defined by $\{(x, y): 2x - y = 10\}$, then R is

(a) Reflexive (b) Symmetric

(c) Transitive (d) None of the above

12. If the function $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in every interval, then

(a) $k < 3$ (b) $k \leq 3$

(c) $k > 3$ (d) $k \geq 3$

13. If $f(x) = \begin{cases} \frac{3\sin \pi x}{5x}, & x \neq 0 \\ 2k, & x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

(a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{10}$

(c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{5}$

14. The domain of the function $f(x) = \sqrt{\cos x}$ is

(a) $\left[\frac{3\pi}{2}, 2\pi\right]$ (b) $\left[0, \frac{\pi}{2}\right]$

(c) $[-\pi, \pi]$ (d) $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$

15. If $f(x) = \begin{cases} \frac{\log x}{x-1}, & \text{if } x \neq 1 \\ k, & \text{if } x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

16. In the given figure, the shaded regions are bounded by straight lines and vertical boundaries. Which of the following statements correctly describes them?

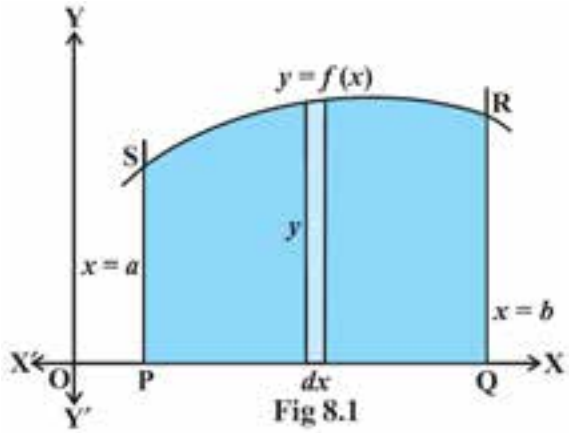


Fig 8.1

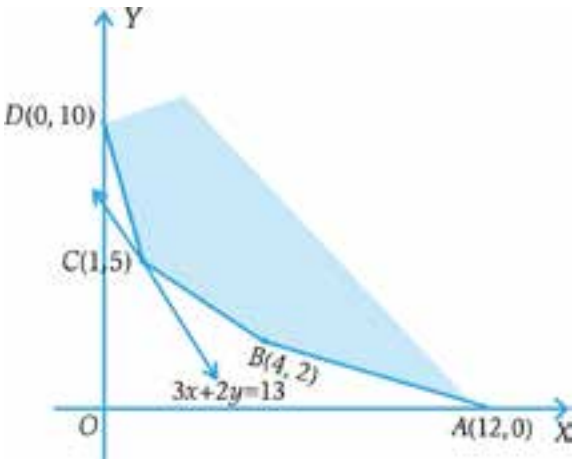
(a) Left shaded region lies between x=-1 and x=-2/3, bounded by line segments AB and AC.

(b) Right shaded region lies between x=0 and x=1, bounded by line segments OD and OE.

(c) Both shaded regions together form trapeziums.

(d) Both (a) and (b) are correct.

17. With the same feasible region, find the minimum value of $z = 4x + y$.



(a) 48 (b) 18

(c) 9 (d) 10

18. Inverse of the matrix $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ is

(a) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ (d) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

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19. Let A be a 2×2 matrix.

Assertion: adj (adj A) $= A$.

Reason: $|\text{adj } A| = |A|$.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** The equation of curve passing through (3, 9) which satisfies differential equation $\frac{dy}{dx} = x + \frac{1}{x^2}$ is $6xy = 3x^3 + 29x - 6$

Reason: The solution of differential equation $\left(\frac{dy}{dx}\right)^2 - \left(\frac{dy}{dx}\right)(e^x + e^{-x}) + 1 = 0$ is $y = c_1 e^x + c_2 e^{-x}$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line has direction ratios $(2, -1, -2)$, then what are its direction cosines?

22. Let R is the equivalence relation in the set $A = \{0, 1, 2, 3, 4, 5\}$ given by $R = \{(a, b): 2 \text{ divides } (a - b)\}$. Write the equivalence class $[0]$.

23. Write the projection of $(\vec{b} + \vec{c})$ on $\vec{a}$, where $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$.

OR

If $\vec{a}$ and $\vec{b}$ are to vectors such that $|\vec{a} + \vec{b}| = |\vec{a}|$, the prove that vector $2\vec{a} + \vec{b}$ is perpendicular to vector $\vec{b}$.

24. Two dice are thrown n times in succession. What is the probability of obtaining a doublet six at-least once?

25. Evaluate $\int \frac{(\log x)^2}{x} dx$.

OR

Evaluate $\int \frac{e^{\tan^{-1} x}}{1 + x^2} dx$.

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. If $x = \cos t(3 - 2\cos^2 t)$ and $y = \sin t(3 - 2\sin^2 t)$, then find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$.

27. If $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}| = 5$, $|\vec{b}| = 6$ and $|\vec{c}| = 9$, then find the angle between $\vec{a}$ and $\vec{b}$.

28. Solve the following equation for x.

$$\cos(\tan^{-1} x) = \sin\left(\cot^{-1} \frac{3}{4}\right)$$

29. Find $|AB|$, if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$.

30. Find the intervals in which the function given by $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ is

(i) increasing

(ii) decreasing.

OR

Show that of all the rectangles of given area, the square has the smallest perimeter.

31. Evaluate $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + e^x} dx$.

OR

Find $\int \frac{x^3}{x^4 + 3x^2 + 2} dx$.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$.

33. Maximise $Z = 5x + 3y$ subject to the constraints: $3x + 5y \le 15$; $5x + 2y \le 10$, $x \ge 0$, $y \ge 0$.

OR

Minimize $Z = 3x + 5y$ such that $x + 3y \ge 3$, $x + y \ge 2$, x, $y \ge 0$.

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34. Find the particular solution of the differential equation

$$x\frac{dy}{dx} - y + x\,\text{cosec}\left(\frac{y}{x}\right) = 0$$

OR

Find the particular solution of the differential equation

$$\frac{dy}{dx} = 1 + x + y + xy,$$

given that $y = 0$ when $x = 1$.

35. Find the vector and Cartesian equations of the line passing through the point (2,1,3) and perpendicular to the lines

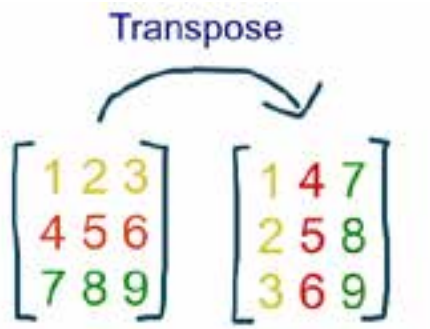
$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

and $$\frac{x}{-3} = \frac{y}{2} = \frac{z}{5}.$$

SECTION - E

Case study based questions are compulsory.

36. In linear algebra, the transpose of a matrix is an operator which flips a matrix over its diagonal; that is, it switches the row and column indices of the matrix A by producing another matrix, often denoted by A^T. The transpose of a matrix was introduced in 1858 by the British mathematician Arthur Cayley.



If $A = [a_{ij}]$ be a $m \times n$ matrix, then the matrix obtained by interchanging the rows and columns of A is called the transpose of A. A square matrix $A = [a_{ij}]$ is said to be symmetric, $A^T = A$ for all possible values of i and j. A square matrix A $[a_{ij}]$ is said to be skew-symmetric, if $A^T = -A$ for all possible values of i and j. Based on the above, information, answer the following questions.

(i) Find the transpose of [1, -2, -5].

(ii) Find the transpose of matrix (ABC).

(iii) Evaluate $(A+B)^T - A$, $A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

OR

Evaluate $(AB)^T$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

37. RK Verma is production analysts of a ready-made garment company. He has to maximize the profit of company using data available. He find that $P(x) = -6x^2 + 120x + 25000$ (in Rupee) is the total profit function of a company where x denotes the production of the company.



Based on the above information, answer the following questions.

(i) Find the profit of the company, when the production is 3 units.

(ii) Find $P'(5)$

(iii) Find the interval in which the profit is strictly increasing.

OR

Find the production, when the profit is maximum.

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38. Measures of joint and survivor life expectancy are potentially useful to those designing or evaluating policies affecting older couples and to couples making retirement, savings, and long-term care decisions. However, couple-based measure of life expectancies are virtually unknown in the social sciences.



The odds against a husband who is 45 yr old, living till he is 70 are. 7 : 5 and the odds against his wife who is now 36, living till she is 61 are 5 : 3.

On the basis of above information, answer the following questions.

(i) Find the probabilities of husband living till 70 and wife living till 61.

(ii) Find the probability P(couple will be alive 25 yr hence).

(iii) Find the probability P(exactly one of them will be alive 25 yr hence).

OR

Find the probability P(none of them will be alive 25 yr hence) and probability P(at-least one of them will be alive 25 yr hence).

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